

FREISTADT, Austria

WMO: 110150

Lat: 48.507N

Lon: 14.507E

Elev: 1798

StdP: 13.77

Time Zone: 1.00 (EUC)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	10.4	1.8	6.4	5.4	7.2	8.5	11.1	12.2	35.1	10.3	35.8	1.9	10	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	85.2	68.4	81.6	66.4	78.4	64.5	69.4	82.4	67.6	79.2	65.8	75.7	3.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.0	98.8	74.6	63.4	93.4	72.4	62.0	88.8	70.7	34.5	82.6	33.0	79.5	31.6	75.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.4	7.2	-0.7	90.8	5.9	2.7	-5.0	92.8	-8.5	94.4	-11.8	96.0	-16.1	98.0		
(5)			-1.1	72.4	5.7	2.0	-5.1	73.9	-8.4	75.0	-11.6	76.1	-15.7	77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	26.9	30.9	37.6	46.9	55.0	60.8	63.9	63.4	55.0	46.6	37.3	30.0		
	DBStd	14.68	8.31	8.16	7.50	7.52	6.52	6.73	5.79	5.65	5.92	6.85	6.86	7.41		
	HDD50	2979	716	535	386	143	26	3	0	1	19	148	382	620		
	HDD65	6980	1181	955	849	544	313	162	92	96	303	570	830	1085		
	CDD50	1620	0	0	2	49	182	326	430	417	169	43	2	0		
	CDD65	146	0	0	0	0	4	35	57	47	3	0	0	0		
	CDH74	1851	0	0	0	14	112	456	657	557	54	1	0	0		
	CDH80	460	0	0	0	1	10	121	185	139	4	0	0	0		
	WSAvg	2.7	2.7	2.9	3.3	3.3	3.1	2.9	2.8	2.7	2.4	2.4	2.3	2.3		
	PrecAvg	31.6	1.7	1.6	2.2	2.0	3.2	4.0	4.4	4.1	2.6	1.9	1.9	1.9		
(15) Precipitation	PrecMax	47.3	3.2	3.8	4.8	3.7	7.0	7.8	8.1	15.7	6.9	4.6	3.4	3.3		
	PrecMin	25.1	0.3	0.4	0.6	0.4	0.6	1.7	2.2	1.9	0.6	0.4	0.2	0.4		
	PrecStd	4.6	0.9	0.9	1.1	1.0	1.3	1.5	1.4	2.7	1.3	1.2	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	54.5	64.3	75.6	81.4	87.8	89.7	89.3	79.1	70.9	61.1	49.5		
		MCWB	44.3	45.7	50.4	58.1	64.1	69.3	70.6	69.6	64.5	59.5	52.6	45.1		
	2%	DB	45.4	49.1	59.1	70.4	77.4	83.9	84.9	84.0	75.0	65.5	54.5	45.4		
		MCWB	41.7	43.1	48.7	56.0	62.1	67.7	68.2	68.2	62.9	56.7	48.8	42.7		
	5%	DB	41.8	45.4	54.3	65.8	73.8	79.5	81.3	79.9	71.1	61.6	50.4	42.1		
		MCWB	39.3	40.7	46.3	53.5	60.1	65.2	66.5	65.6	60.8	54.7	46.9	39.8		
	10%	DB	38.5	42.0	49.7	61.7	69.7	75.2	77.2	76.6	67.2	58.1	47.1	39.3		
		MCWB	36.3	38.6	44.0	51.3	58.2	62.9	64.7	64.1	58.9	52.9	44.4	37.4		
	0.4%	WB	45.0	46.9	51.8	59.3	65.5	71.1	71.8	72.1	66.2	60.2	52.8	45.7		
		MCDB	48.5	52.7	61.4	73.7	78.6	84.6	86.6	86.2	76.6	68.4	59.9	47.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.1	43.6	49.5	56.5	63.2	68.6	69.3	68.8	63.6	57.6	49.5	42.7		
		MCDB	45.2	47.9	57.5	68.6	75.2	81.5	82.1	81.6	72.6	64.0	53.2	45.1		
	5%	WB	39.4	41.2	47.0	54.2	61.0	66.3	67.6	66.8	61.6	55.5	47.1	40.0		
		MCDB	41.7	45.0	53.2	64.2	71.8	77.3	78.8	77.6	69.3	60.4	50.2	42.1		
	10%	WB	36.5	38.7	44.7	52.1	58.9	64.3	65.7	65.0	59.7	53.4	44.6	37.5		
		MCDB	38.3	41.8	49.4	60.6	68.2	73.0	75.1	74.2	66.1	57.4	47.0	39.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	14.2	16.5	19.7	19.9	19.7	19.6	19.9	18.4	17.7	12.4	10.7		
		MCDBR	14.4	20.0	26.5	28.7	29.0	28.4	28.3	27.7	25.8	23.5	19.4	14.8		
		MCWBR	11.4	14.1	16.3	15.3	14.5	13.0	12.5	12.7	14.0	15.0	13.6	11.7		
	5% WB	MCDBR	13.2	18.1	24.4	26.5	26.6	25.7	25.1	25.6	23.2	21.3	18.2	13.6		
		MCWBR	10.6	13.0	15.6	14.6	13.7	12.2	11.5	12.4	13.1	14.2	13.2	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.314	0.353	0.390	0.395	0.398	0.406	0.401	0.374	0.344	0.318	0.290		
	taud		2.390	2.356	2.319	2.275	2.297	2.318	2.295	2.326	2.373	2.440	2.465	2.447		
	Ebn at Noon		248	266	272	272	275	274	270	266	262	252	235	233		
	Edh at Noon		21	28	34	39	40	39	40	37	32	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		316	597	933	1368	1582	1728	1669	1488	1061	659	334	249		
	RadStd		51	79	120	168	184	154	162	150	137	76	35	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (96 neighbors)	+0.90	N/A	N/A	N/A	+0.70	+0.61	N/A	N/A	+122	+54

Nomenclature: See separate page